

Clinical Reasoning Manual

NBEO Part II PAM / TMOD. Illness scripts, an emergency gate, treatment grammar, and two 175-item case simulations with a balanced answer key.

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350	175	3.5	~120
ITEMS, IN CASES OF 3-6	ITEMS PER SESSION	HOURS PER SESSION	TMOD ITEMS EMBEDDED

WHAT CHANGED IN THIS EDITION

The previous edition's answer key was 82% option A, and 95% in Session 2. Every key has been rebuilt to an even letter spread, every item now carries four credible options, and every rationale explains the tempting wrong answer as well as the right one.

FORMAT FACTS, VERIFIED

Item count, case size, session structure, timing, and the separate TMOD breakout decision were checked against NBEO's published exam page on 15 August 2026. Domain and item-type figures are transcribed and marked unconfirmed on page 2.

What is checked, and what is not

A guide that labels its own uncertainty is more useful than one that asserts it was verified. Every factual claim carries one of three states.

Verification table

CLAIM	STATE	BASIS
350 items, presented as cases of 3-6 items	VERIFIED	NBEO exam page, fetched 15 Aug 2026
Two sessions of 175 items, 3.5 hours each	VERIFIED	NBEO exam page
Approximately 120 TMOD items, separate breakout decision	VERIFIED	NBEO exam page
Delivered at Pearson centres; tutorial being updated	VERIFIED	NBEO exam page
Per-domain item ranges, 17 domains	UNCONFIRMED	Transcribed from prior edition. Content Matrix PDF resolves but was not machine-read
Item-type percentage bands	UNCONFIRMED	As above
Deep link to the Pearson tutorial	REMOVED	URL did not resolve; reach it from the exam page

The defect this edition exists to fix

ANSWER KEY	SESSION 1	SESSION 2	BOTH
Single-answer items	161	172	333
Correct answer was A	67.1%	95.3%	81.7%
Longest run of one letter	30	140	-
Letters never correct	E	D, E	E

WHAT THAT MEANT IN PRACTICE

Answering A to every single-answer item scored 272/333, about 82%, without reading one stem. Session 2 alone scored 95%. The two sessions were also biased by different amounts, so a Session 1 score could not be compared with a Session 2 score, which broke the guide's own repair-and-confirm workflow.

REBUILD RULE, AUDITABLE

Each letter A to E is correct on 18-22% of single-answer items in each session, no letter repeats more than three times consecutively, all five letters appear in both sessions, and option length is matched within each item. Re-run tools/key_audit.py after any edit to the bank.

Six passes

Reading this manual is not studying it. Retrieval under time pressure is what transfers to the exam room.

PASS	WORK	PROOF IT WORKED
1 • Systems	Learn the algorithm, the emergency gate, the condition cards, the drug anchors.	You can state recognition, danger, next test, treatment, contraindication, follow-up, and failure trigger without looking.
2 • Retrieval	Closed-book recall at roughly 1, 3, 7, 14, and 30 days, always from a changed prompt.	Successful recalls are logged. Hours are not.
3 • Session 1	175 items, 210 minutes, uninterrupted, confidence marked on every item.	Scored by item type as well as total, with pacing and error categories written down.
4 • Repair	High-confidence misses first. Each becomes one if-then rule.	Every false rule has a replacement you can recite.
5 • Session 2	Second 175 items under identical conditions.	Weak item types improve; late-session accuracy holds.
6 • Taper	Red flags, drugs, formulas, image algorithms, logistics.	No new content in the final 48 hours. Sleep protected.

Error log: seven fields, one line each

Concept	Short disease or decision label.
Answer contrast	What you chose against what was correct.
Error type	Knowledge, discrimination, image reading, management sequence, calculation, or misreading the stem.
False rule	The exact belief that produced the miss, written as a sentence.
New rule	One executable if-then replacement.
Next recall	Tomorrow for a high-confidence miss, three days for low-confidence, seven days after a success.
Transfer test	A changed patient or the competing diagnosis.

WHY SCORE BY ITEM TYPE

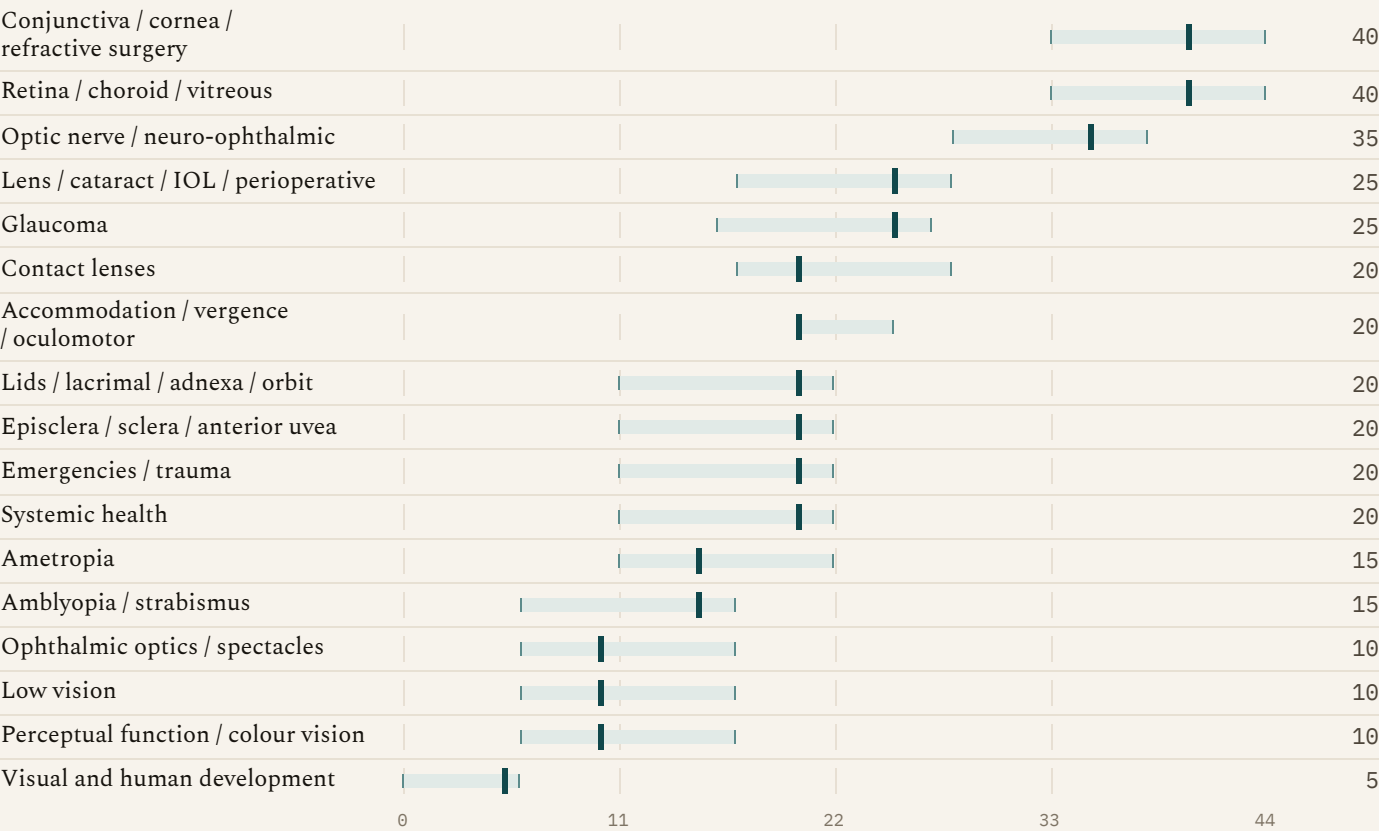
The blueprint weights treatment and management far above basic science, and TMOD is decided separately. A 74% total can hide a failing treatment score. Every item here is tagged, so scoring by tag takes two minutes and tells you what to repair.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital. The exam tests the discriminator, never the textbook description.

Know the target

Band shows the published range. Bar shows this manual's 350-item allocation. Scale runs 0 to 44 items.



READ IT THIS WAY

Cornea and retina alone can carry 88 items. Revision split evenly across 17 domains is misallocated by a factor of eight at the extremes. Development items usually arrive inside paediatric cases.

Case conventions worth memorising

BVA	Best achievable or best-corrected acuity. If it is reduced, pinhole improvement has already been accounted for, so do not spend an item hunting refractive causes.
Near acuity	Assume 16 inches unless the case says otherwise.
Fields	Side by side: OD right, OS left. Stacked: OD above OS.
Images	Fundus, OCT, fields, angiography, ultrasound, radiology, anterior segment. Describe before diagnosing.

245	105
ITEMS · NORMAL HEALTH, DISEASE, TRAUMA	ITEMS · REFRACTIVE, SENSORY, OCULOMOTOR

Where the marks are

Item-type weighting and this manual's allocation

ITEM TYPE	BAND	HERE	WHAT IT ACTUALLY ASKS
Treatment / management	40-57%	163 · 47%	Choice, mechanism, dose, schedule, duration, follow-up interval, education, prognosis, escalation.
Diagnosis	26-34%	113 · 32%	Most likely diagnosis, the supporting datum, the missing datum, the next diagnostic test.
Basic science, applied	11-17%	59 · 17%	Anatomy, physiology, pathophysiology, microbiology, optics, pharmacology, tied to this patient.
Legal / ethics / public health	4-10%	15 · 4%	Duty, consent, refusal, records, delegation, telehealth, epidemiology, screening arithmetic.

Bands transcribed from the prior edition and marked unconfirmed on page 2. Allocation sums to 350.

Pacing

CASE	BUDGET	MOVE	Multiple-response discipline	ALL OR NONE
3 items	4 min	Read once, answer the direct items, flag one uncertainty.	COUNT	Underline the requested number before reading options.
4 items	5 min	Commit before option-hunting for rescue.	JUDGE	Each option alone, true or false for this patient.
5 items	6 min	The typical unit. Do not reread the stem per item.	PREFER	Patient-specific over merely true.
6 items	7 min	Diagnosis, threat, test, treatment, follow-up.	SCORE	Every correct choice and no incorrect choice. A partial set earns zero.
			TRAP	Findings that prove the diagnosis are not the same as findings that measure organ function. Read which the item wants.

72 seconds per item. Checkpoints: item 50 by 60 minutes, item 100 by 120, item 150 by 180. Behind schedule means cut rereading, not standards.

Seven-step case algorithm

1 Frame	Age, laterality, tempo, pain, vision loss, systemic context, medications, risk behaviour.
2 Threat gate	Could delay cost life, eye, or fellow eye? Run the twelve patterns on pages 6 and 7.
3 Localise	Media, cornea and anterior segment, retina, optic nerve, post-chiasmal pathway, neuromuscular junction, orbit, binocular motor system.
4 Build three	Most likely, most dangerous plausible, best mimic.
5 Discriminate	The one finding or test that most changes probability or management. Not a panel.
6 Treat in order	Stabilise, protect tissue and vision, treat cause, prevent recurrence and fellow-eye injury.
7 Close the loop	Interval, measurable response, adverse effects, red flags, referral urgency, and what failure triggers.

Six patterns that cannot wait

Recognise, act, and know the specific thing that must not be done. The prohibition is examined as often as the treatment.

Chemical injury SURFACE	Open globe TRAUMA
RECOGNISE Exposure history. Do not wait for acuity or pH before acting.	RECOGNISE Full-thickness wound, Seidel, peaked pupil, uveal prolapse, low IOP with a mechanism.
DO FIRST Irrigate immediately and copiously. Evert lids, sweep fornices for retained particles, continue until pH is neutral and stays neutral after a pause. Then grade limbal ischaemia and epithelial loss.	DO FIRST Rigid shield, nil by mouth, antiemetic and analgesia, tetanus and systemic antibiotics per protocol, CT orbits when appropriate, immediate surgical referral.
NEVER Neutralise alkali with acid. Delay irrigation to find equipment.	NEVER Tonometry, gonioscopy, ultrasound, or any pressure on the globe.
Orbital cellulitis ORBIT	Microbial keratitis CORNEA
RECOGNISE Lid swelling plus any of: painful or restricted motility, proptosis, reduced acuity or colour, RAPD, raised IOP, fever, severe chemosis.	RECOGNISE Contact lens wear, epithelial defect over a stromal infiltrate, chamber reaction, pain, central location.
DO FIRST Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ophthalmology and ENT. Drain a subperiosteal abscess when vision or the apex is threatened.	DO FIRST Measure and photograph. Culture central, large, deep, atypical, post-surgical, or unresponsive ulcers. Intensive topical antimicrobial, review inside 24 hours.
NEVER Call it preseptal until orbital signs have been actively excluded.	NEVER Send a central ulcer home on conjunctivitis dosing, or patch it.
Acute angle closure GLAUCOMA	Giant cell arteritis SYSTEMIC
RECOGNISE Pain, halos, nausea, corneal oedema, shallow chamber, mid-dilated fixed pupil, very high IOP.	RECOGNISE Over 50 with new headache, jaw claudication, scalp tenderness, polymyalgia or constitutional symptoms, chalky pallid disc oedema.
DO FIRST Aqueous suppressants the patient can safely take, systemic acetazolamide if appropriate, control inflammation and nausea, then definitive iridotomy. Assess the fellow eye.	DO FIRST ESR, CRP, and platelets, and start high-dose systemic corticosteroid immediately. Temporal artery imaging or biopsy follows treatment.
NEVER Lead with pilocarpine at extreme pressure: an ischaemic sphincter will not respond.	NEVER Let a normal ESR reassure you, or wait for biopsy. The fellow eye is what is at stake.

CONTINUED ON PAGE 7

Retinal artery occlusion, tear and detachment, endophthalmitis, papilloedema, acute third-nerve palsy, painful Horner syndrome, and the treatment grammar that every management answer has to satisfy.

And six more

Retinal artery occlusion VASCULAR RECOGNISE Sudden painless profound monocular loss, dense RAPD, retinal whitening or cherry-red spot. DO FIRST Treat as acute stroke: activate the emergency stroke pathway now, with carotid and cardiac source evaluation and a GCA screen by age. NEVER Spend the treatment window on ocular massage or other office manoeuvres.	Tear and detachment RETINA RECOGNISE New flashes, shower of floaters, pigment in the vitreous, vitreous haemorrhage, curtain or field loss. DO FIRST Dilated peripheral examination with indentation, B-scan if the view is blocked, urgent retina referral. Macula-on detachment is the most time-critical. NEVER Accept a normal central OCT as evidence against a peripheral break.
Endophthalmitis POSTOPERATIVE RECOGNISE Pain and severe acuity loss after surgery or injection, hypopyon, fibrin, vitritis, lost red reflex. DO FIRST Same-hour retina-level care: intraocular sampling then intravitreal antibiotics, with vitrectomy by severity. NEVER Treat it as routine postoperative inflammation, or wait for culture before treating.	Papilloedema NEURO RECOGNISE Bilateral disc oedema with headache, transient obscurations, pulsatile tinnitus, diplopia, possible sixth-nerve palsy. DO FIRST Urgent brain imaging including venous evaluation, then lumbar puncture with opening pressure when imaging makes it safe. NEVER Puncture before imaging, or dismiss true oedema as crowded discs without evidence.
Third-nerve palsy NEURO-VASCULAR RECOGNISE Ptosis with the eye down and out; pupil involvement, pain, or an incomplete palsy raises compression. DO FIRST Emergency aneurysm-focused CTA or MRA and neurovascular referral. Modern practice images acute isolated palsies. NEVER Treat pupil sparing as proof against aneurysm in an acute palsy.	Painful Horner syndrome NEURO-VASCULAR RECOGNISE Acute ptosis and miosis, anisocoria greater in darkness, dilation lag, with head or neck pain. DO FIRST Emergency head and neck vascular imaging for carotid dissection. NEVER Delay imaging to complete pharmacologic localisation.

TREATMENT GRAMMAR, EVERY TIME

Immediate action, drug or procedure, dose and form and schedule and duration, contraindication, patient education, follow-up interval, the measurable response, and the escalation trigger. An answer that names a drug but no interval or failure trigger is incomplete, and incomplete options are how these items are written.

Ocular surface and adnexa

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Preseptal cellulitis LIDS / ORBIT PATTERN Lid erythema and oedema with normal acuity, pupils, colour, motility, and globe position. CONFIRM Clinical. Image when the examination is limited, severe, atypical, or orbital spread is possible. ACT Oral antibiotic covering skin and sinus organisms, review inside 24-48 hours, written orbital red flags. TRAP In a febrile child with sinusitis this is a diagnosis of exclusion. Motility and colour are the gate.	Dacryocystitis LACRIMAL PATTERN Tender erythematous swelling below the medial canthal tendon, with reflux or tearing. CONFIRM Clinical, culture the drainage, assess for orbital extension. ACT Systemic antibiotics, warm compresses, drainage of an abscess, definitive lacrimal surgery once settled. TRAP Do not probe or irrigate through acutely infected tissue.
Recurrent chalazion LIDS PATTERN Firm lesion returning to the same site, madarosis, yellow tarsal thickening, unilateral chronic blepharoconjunctivitis. CONFIRM Full-thickness biopsy with conjunctival mapping, not a third curettage. ACT Urgent oculoplastic referral. Evert both lids, examine conjunctiva and regional nodes. TRAP Sebaceous carcinoma masquerades as chalazion and as blepharitis, and spreads pagetoid.	Thyroid eye disease ORBIT PATTERN Lid retraction, proptosis, exposure, restrictive diplopia, apical crowding. CONFIRM Thyroid function, colour and pupil testing, fields, exophthalmometry; imaging when atypical or neuropathy is suspected. ACT Surface protection, smoking cessation, urgent specialist care for optic neuropathy or corneal breakdown. TRAP Activity and severity are separate axes. Muscle belly enlargement with tendon sparing is the signature.
Adenoviral keratoconjunctivitis CONJUNCTIVA PATTERN Follicles, watery discharge, preauricular node, high contagion; subepithelial infiltrates may follow. CONFIRM Clinical; rapid antigen testing helps in selected settings. ACT Supportive care and strict hygiene. Corticosteroid only for visually significant infiltrates, monitored. TRAP Contagious period and instrument disinfection matter as much as the drop.	Episcleritis vs scleritis EPISCLERA / SCLERA PATTERN Sectoral mobile vessels that blanch with phenylephrine, against deep violaceous non-blanching injection with boring pain and globe tenderness. CONFIRM Phenylephrine, palpation, B-scan for posterior disease, phenotype-directed systemic workup. ACT Episcleritis: lubricant, cool compress, NSAID. Scleritis: systemic anti-inflammatory or immunomodulatory therapy, urgent specialist. TRAP Topical steroid alone is inadequate in scleritis and dangerous if the cause is infectious.

Anchor, caveat, monitor

A memorised dose used in the wrong tissue or the wrong host is a wrong answer.

Antimicrobial anchors

INDICATION	ANCHOR	CAVEAT THAT GETS EXAMINED
Bacterial keratitis	Topical fluoroquinolone for small peripheral ulcers; fortified therapy for severe, central, deep, or resistant disease. Load every 5-15 minutes, then hourly.	Culture first when central, large, deep, atypical, or unresponsive. Do not wait for results to treat.
HSV epithelial keratitis	Oral aciclovir 400 mg five times daily, or valaciclovir 500 mg three times daily, 7-10 days. Topical ganciclovir 0.15% is an alternative.	No steroid over an active epithelial ulcer. Steroid belongs to immune stromal or endothelial disease, with antiviral cover.
Herpes zoster ophthalmicus	Valaciclovir 1 g three times daily, or aciclovir 800 mg five times daily, 7-10 days, renally adjusted.	Ocular disease lags the rash. A quiet first examination does not end surveillance.
Fungal keratitis	Natamycin 5% hourly initially for filamentous organisms, prolonged course, cornea specialist involved.	Corticosteroid accelerates progression. Vegetative trauma and feathery margins are the clues.
Acanthamoeba keratitis	Intensive biguanide therapy, PHMB 0.02% or chlorhexidine 0.02%, often with a diamidine, for months.	Pain out of proportion, water exposure. Early disease is mislabelled HSV.
Hyperacute conjunctivitis	Systemic ceftriaxone-based treatment per current CDC guidance, saline lavage, STI testing, partner management.	Copious purulence with corneal risk is an emergency, not a drop problem.

Contraindication anchors

Topical beta-blocker	CAUTION	Alpha-2 agonist	CAUTION
AVOID Bronchospastic asthma or COPD, bradycardia, heart block, decompensated heart failure.		AVOID Brimonidine in infants and very young children: CNS depression and apnoea.	
REDUCE Punctal occlusion and lid closure lower systemic absorption.		EXPECT Fatigue, dry mouth, late follicular allergy, hypotension.	
TRAP A resting pulse in the low 50s is in the stem for a reason.		TRAP Check interacting systemic drugs before prescribing.	
Miotic	CAUTION	Topical corticosteroid	CAUTION
AVOID Susceptible peripheral retina, active inflammation, ciliochoroidal effusion mechanisms.		AVOID Active HSV epithelial disease, fungal and Acanthamoeba keratitis, uncontrolled bacterial infection.	
EXPECT Brow ache, induced myopia, poor night vision.		MONITOR IOP, epithelial healing, cataract, and the taper.	
TRAP Useless with an ischaemic sphincter, harmful in topiramate-type effusion.		TRAP Name tissue, cause, dose, and monitoring, or the answer is incomplete.	

Arithmetic, fast

The fastest marks on the exam, and the easiest to lose to a sign error.

Formulas

NAME	FORM	USE AND SIGN DISCIPLINE
Spherical equivalent	sph + cyl/2	Halve the cylinder with its own sign. -6.00 -2.00 x 180 gives -7.00 D.
Transposition	sph + cyl, flip, 90°	New sphere is old sphere plus old cylinder, reverse the cylinder sign, rotate the axis by 90 degrees.
Prentice rule	$\Delta = \text{cm} \times \text{D}$	Decentration in centimetres. Base follows the thick edge for plus lenses, the thin edge for minus.
Vertex conversion	$F2 = F1/(1 - dF1)$	Distance in metres, signs preserved. Matters beyond about ± 4.00 D.
Kestenbaum estimate	1 / acuity fraction	Starting magnification only. Refine by task, working distance, contrast, field, endurance.
Equivalent viewing power	1 / metres	10 cm working distance is 10 D.
Hofstetter	$18.5 - 0.30 \text{ age}$	Average. Minimum is $15 - 0.25 \text{ age}$, maximum $25 - 0.40 \text{ age}$. A screening comparison, not a diagnosis.
Gradient AC/A	$\Delta \text{phoria} / \Delta \text{D}$	State the sign convention before you calculate.
Screening arithmetic	$\text{TP}/(\text{TP}+\text{FN})$	Sensitivity. Specificity is $\text{TN}/(\text{TN}+\text{FP})$. PPV rises with prevalence; NPV falls.

Drills: cover the right column

PROMPT	WORKING AND ANSWER
Transpose +2.00 -3.00 x 180	-1.00 +3.00 x 090
Spherical equivalent of -4.00 -2.00 x 090	-4.00 + (-1.00) = -5.00 D
4 mm decentration through +5.00 D	$0.4 \times 5 = 2\Delta$
-10.00 D spectacle at 12 mm, to the corneal plane	$-10/(1 - (0.012 \times -10)) = -8.93 \text{ D}$
Starting magnification for 20/160	$160/20 = 8\times$
Hofstetter average amplitude at age 20	$18.5 - 6.0 = 12.5 \text{ D}$
Hydroxychloroquine 400 mg daily at 60 kg	6.67 mg/kg, above the 5 mg/kg threshold
Soft toric rotated 10° left of a 180 axis	LARS: order 010
GP steepened 0.50 D, over-refraction -0.50 D, from -3.00 D	SAM: $-3.00 -0.50 -0.50 = -4.00 \text{ D}$
Anisometropia -2.00 OD, -5.00 OS, 10 mm below centre	2Δ and 5Δ base down: 3Δ imbalance

Marks are lost to sign errors in transposition and vertex, to forgetting that Prentice needs centimetres, and to quoting a magnification estimate as a prescription. Write one line of working every time.

CASE S1-01 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

The swollen eyelid that hurts to move

A 13-year-old boy has three days of worsening left eyelid swelling and frontal headache, after a week of purulent nasal discharge. Temperature 38.4 °C. BVA 20/20 OD, 20/40 OS. Mild left RAPD, red saturation 60% OS, 3 mm proptosis, tense chemosis, and painful restriction of elevation and abduction. IOP 17 OD, 27 OS. Cornea clear, chamber quiet, fundus normal both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Preseptal cellulitis with reactive chemosis
- B Idiopathic orbital inflammatory syndrome
- C Cavernous sinus thrombosis complicating sinusitis
- D Orbital cellulitis of ethmoid sinus origin
- E Orbital rhabdomyosarcoma with secondary inflammation

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings measure optic nerve or orbital function, as opposed to supporting the infective diagnosis?

SELECT 3

- A Temperature 38.4 °C
- B Left RAPD
- C Red saturation reduced to 60% OS
- D Purulent nasal discharge for one week
- E IOP 27 mmHg OS
- F BVA 20/40 OS

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Oral amoxicillin-clavulanate, nasal decongestant, and review in 24 hours
- B Same-day emergency referral: contrast CT of orbits and sinuses, intravenous antibiotics, ophthalmology and ENT
- C Outpatient MRI within 48 hours, then decide on admission
- D Intravenous antibiotics arranged for the following morning, with oral cover overnight
- E Oral corticosteroid with topical antibiotic to settle the orbital swelling

ANSWER _____ CONFIDENCE L • M • H

4 Which anatomic route most commonly explains this presentation?

BASIC SCIENCE

- A Direct spread across the lamina papyracea from the ethmoid sinus
- B Retrograde thrombophlebitis through valveless facial and ophthalmic veins
- C Extension through the optic canal from the sphenoid sinus
- D Ascending infection through the nasolacrimal duct from the inferior meatus
- E Haematogenous seeding from occult bacteraemia

ANSWER _____ CONFIDENCE L • M • H

5 He is admitted on intravenous antibiotics. Which change over 18 hours most supports surgical drainage?

TREATMENT

- A Temperature settles after two doses while lid oedema persists
- B CT shows a 4 mm medial subperiosteal collection with acuity and motility improving
- C Acuity falls to 20/80 with worsening colour and an abscess crowding the orbital apex
- D Blood cultures grow a fully sensitive organism
- E Chemosis increases while pupils, colour, and acuity stay stable

ANSWER _____ CONFIDENCE L • M • H

Case Sl-01 • why

Review order: wrong answers, then low-confidence correct answers, then high-confidence misses.

1 • D Orbital cellulitis of ethmoid sinus origin

Proptosis, painful restricted motility, RAPD, reduced colour and acuity, and raised IOP place the disease behind the septum. Fever with a week of purulent rhinorrhoea in a child names the source.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Cavernous sinus thrombosis	It is the correct escalation of this exact infection, and it also produces proptosis, chemosis, and reduced vision.	Signs are strictly unilateral, with no multiple cranial neuropathy, no V1-V2 sensory loss, and no bilateral progression.
A	Preseptal cellulitis	The visible finding is a swollen lid, and preseptal disease is far more common at this age.	Preseptal disease preserves acuity, pupils, colour, motility, and globe position. Four of the five are abnormal here.
E	Rhabdomyosarcoma	Rapid proptosis in a child is the classic teaching phrase for it.	Fever, purulent sinus disease, and a three-day course fit infection; tumour proptosis is not febrile and not this fast.

2 • B, C, F RAPD, red desaturation, reduced BVA

The item asks for afferent function, not for evidence of infection. RAPD, dyschromatopsia, and reduced best acuity each measure optic nerve conduction directly.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	IOP 27 mmHg	It is genuinely abnormal, and raised orbital pressure is the thing threatening the nerve.	It measures orbital pressure, not nerve function. Normalising it would not tell you the nerve had recovered.
A, D	Fever, purulent discharge	They confirm infection and drive the diagnosis.	Neither measures the eye. A child can be febrile with an entirely normal afferent system.

3 • B Same-day emergency referral, imaging, intravenous antibiotics

Orbital cellulitis with afferent dysfunction threatens the eye, and through intracranial extension, life. Imaging, intravenous therapy, and surgical availability have to be arranged together, today.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Intravenous antibiotics tomorrow morning	It correctly identifies that intravenous therapy is needed and feels like a workable compromise with access.	The overnight delay is where vision is lost. The decision point is the RAPD, and it exists now.
C	Outpatient MRI in 48 hours	MRI is the better soft-tissue modality, so it reads as the more thorough answer.	Contrast CT answers the urgent questions faster: sinus disease, subperiosteal collection, apical crowding, bone.

Case Sl-01 • why, continued

4 • A Lamina papyracea from the ethmoid sinus

The medial orbital wall is paper-thin and perforated by vessels, which makes contiguous ethmoid spread the usual route and explains why abduction and elevation are restricted first.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Valveless venous spread	It is a real mechanism, and it is how mid-face infection reaches the cavernous sinus.	It is not the common route from paranasal sinusitis and does not explain a medially localised collection.
C	Optic canal from the sphenoid	Sphenoid disease is genuinely dangerous and sits next to the apex.	Sphenoid involvement is much less common in children and would present with early apical signs rather than medial restriction.

5 • C Falling acuity and colour with an abscess crowding the apex

Drainage is driven by function and location, not by the presence of a collection. Deteriorating afferent function with apical involvement is the indication; so is failure to improve on adequate therapy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	4 mm collection, function improving	An abscess on imaging feels like a surgical finding by definition.	Small medial collections in children often resolve on antibiotics while function improves. Imaging alone does not decide.
E	Increasing chemosis, stable function	It looks like clinical deterioration.	Surface swelling without afferent change is not a reason to open an orbit.
A	Fever settles, lid oedema persists	Persistent swelling feels like treatment failure.	Defervescence with stable function is the expected early response. Lid oedema lags by days.

IF-THEN RULES FROM THIS CASE

If lid swelling comes with any abnormality of acuity, colour, pupil, motility, or globe position, then it is orbital until imaging and examination say otherwise. If an abscess is present but function is improving, then continue medical therapy and reassess. If function is falling, then drain, whatever the size on the scan.

ITEM-TYPE SCORE FOR THIS CASE

Two treatment, one diagnosis, one basic science, one multiple-response. Record it that way. Across the session, a treatment score below your diagnosis score is the single most common failing pattern, because treatment items carry the largest share of the blueprint and the TMOD decision.

Cut-line recall deck

Cut on the dashed lines. Ten seconds per prompt, answered aloud.

Misses go back into tomorrow's pile.

Painful red eye in a contact lens wearer Microbial keratitis until the slit lamp says otherwise. Measure it, culture if central, treat intensively, review inside 24 hours. Never patch.	Pain out of proportion, water exposure, poor antibiotic response Acanthamoeba. Culture, PCR, or confocal, repeated if suspicion persists. Ring infiltrate is late, not required. Steroid makes it worse.
Lid swelling plus painful restricted motility Orbital cellulitis. Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ENT. Colour and motility are the gate.	Over 50, profound loss, chalky pallid disc oedema Arteritic AION. ESR, CRP, platelets, and immediate high-dose systemic steroid. A normal ESR does not clear it.
Sudden painless profound monocular loss Retinal artery occlusion, an acute stroke syndrome. Stroke pathway now, embolic source workup, GCA screen by age. No office massage.	Flashes, shower of floaters, pigment in the vitreous Retinal break until the periphery has been examined with indentation. A normal central OCT proves nothing.

Deck, continued

New distortion with subretinal fluid in an older patient Neovascular AMD until proven otherwise. Urgent retina referral for anti-VEGF. Good acuity is a reason to treat quickly, not to wait.	Bilateral myopic shift with uniformly shallow chambers on a new drug Ciliochoroidal effusion, commonly topiramate. Stop the drug with the prescriber, suppress aqueous, cycloplege. Not pupillary block: no iridotomy, no miotics.
Acute ptosis and miosis after neck pain Painful Horner: carotid dissection until imaged. Emergency head and neck vascular imaging. Do not wait to finish localisation drops.	Recurrent chalazion at the same site with lash loss Sebaceous carcinoma. Full-thickness biopsy with conjunctival mapping, examine nodes. Not a third curettage.
Multiple-response item Count the requested number, judge each option alone, choose patient-specific over generally true. Partial sets score zero.	High-confidence miss Top of the error log. Name the false rule, write the replacement as one if-then sentence, retest tomorrow from a changed stem.

FINAL 48 HOURS

Red flags, drug contraindications, formulas, image algorithms, and logistics only. No new content. Between the two sessions, do not autopsy Session 1: eat, hydrate, reset, and start the second session on pacing.

Sources and status

Official NBEO documents control scope and weighting. Everything below was checked on 15 August 2026, and link status is recorded so a reader can tell what was actually opened.

Official

STATUS	DOCUMENT
READ	NBEO. Part II PAM/TMOD exam information. optometry.org/exams/part-ii-pam-tmod — source for item count, case size, session structure, timing, TMOD breakout, and delivery.
RESOLVES	NBEO. Part II Content Matrix, PDF dated May 2026. Link returns 200; content not machine-read for this edition, so domain ranges remain unconfirmed.
RESOLVES	NBEO. Part II PAM Content Outline and TMOD Content Outline, PDFs dated May 2026.
RESOLVES	NBEO. PAM Sample Cases and Test Day Instructions, PDFs dated May 2026.
DEAD	Pearson VUE tutorial deep link. Did not resolve; NBEO states the tutorial is being updated. Reach it from the exam page.

Clinical anchors

Mac Grory B, et al. Management of central retinal artery occlusion. *Stroke* 2021;52:e282-e294. · Marmor MF, et al. Recommendations on screening for chloroquine and hydroxychloroquine retinopathy, 2016 revision. *Ophthalmology* 2016;123:1386-1394. · Herpetic Eye Disease Study Group. *N Engl J Med* 1998;339:300-306. · Beck RW, et al. Corticosteroids in acute optic neuritis. *N Engl J Med* 1992;326:581-588. · AREDS2 Research Group. *JAMA* 2013;309:2005-2015. · CDC. Sexually transmitted infections treatment guidelines, 2021. · American Diabetes Association. Standards of care in diabetes.

Learning method

Dunlosky J, et al. Improving students' learning with effective learning techniques. *Psychol Sci Public Interest* 2013;14:4-58. · Roediger HL, Karpicke JD. Test-enhanced learning. *Psychol Sci* 2006;17:249-255.

KNOWN LIMITS OF THIS EDITION

Domain ranges and item-type bands are transcribed, not confirmed against the current matrix. This specimen contains one rebuilt case, not all seventy. Clinical anchors are exam-oriented simplifications: verify labelling, dose, interactions, and local law before treating a patient.

ORIGINALITY AND INDEPENDENCE

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